

Boa Jang

Medical Foundation Models · Representation Learning · Multimodal Learning

Seoul, Korea · boajang97@gmail.com · +82 10-5601-9547

Google Scholar · github.com/Boa-Jang · linkedin.com/in/boa-jang · boa-jang.github.io

RESEARCH SUMMARY

Ph.D. candidate in Bioengineering and medical AI researcher studying how clinically meaningful representations can be learned from large-scale real-world clinical data. My research spans retinal representation learning, medical foundation models, robustness under cross-device and cross-modality distribution shifts, and multimodal clinical AI. I am particularly interested in understanding how learned representations capture biological changes over time, with a focus on aging, disease progression, and reliable generalization across heterogeneous clinical environments.

EDUCATION

Seoul National University

Ph.D. Candidate in Bioengineering, expected Feb. 2027

Advisors: Jinwook Choi and Young-Gon Kim

Seoul, Korea

Mar. 2023 – Present

Peking University

B.S. in Psychology

Beijing, China

Sep. 2016 – Jul. 2020

RESEARCH EXPERIENCE

Seoul National University Hospital

Graduate Researcher

Seoul, Korea

Mar. 2023 – Present

- Lead independent research across medical foundation models, representation learning, test-time adaptation, and longitudinal analysis for diverse clinical AI problems.
- Develop and analyze models using large-scale real-world clinical datasets, including retinal imaging cohorts with more than 100,000 screening images and longitudinal data from over 20,000 patients.
- Collaborate closely with clinicians throughout the research process, from clinical problem formulation and cohort design to model development, external validation, and expert reader studies.

Medical AI Researcher

Nov. 2021 – Feb. 2023

- Developed and applied machine learning and deep learning models across diverse clinical projects using laboratory data, CT imaging, and digital pathology.

Korea Institute of Science and Technology (KIST)

Research Intern

Seoul, Korea

Aug. 2020 – May 2021

- Studied neural circuits underlying motor control using in vivo behavioral experiments, optogenetics, and fluorescence imaging.

MANUSCRIPTS

[m.2] B. Jang, J. Lee, G. Lee, J. Choi, and Y.-G. Kim. *Skeleton-Guided Progressive Test-Time Adaptation for Thin Curvilinear Structures*. AAAI 2027. **Under review**.

[m.1] B. Woo, Y. Jung, B. Jang, J. Jung, and Y.-G. Kim. *Integrating RNFL and Fundus Photographs for Differentiation of Glaucomatous and Non-Glaucomatous Optic Neuropathy Using Deep Learning*. Scientific Reports. **In revision**.

PEER-REVIEWED PUBLICATIONS

[†] co-first author

[j.10] **B. Jang**[†], R. Oh[†], et al. (2026). Deep Learning-Derived Retinal Age Gap and Its Associations with Lifestyle, Systemic, and Ocular Health in a Health Screening Cohort. *GeroScience*. **Accepted**.

[j.9] B. Woo, J. Choi, ..., **B. Jang**, et al. (2026). Prediction of Postoperative Intracerebral Hemorrhage Following Brain Biopsy Using Integrated Statistical and Machine Learning Analyses. *Scientific Reports*.

[j.8] J.R. Song[†], **B. Jang**[†], et al. (2026). Segmentation-Guided Diffusion Model for Retinal OCT Speckle Reduction with Anatomical Structure Preservation. *Scientific Reports*.

[j.7] **B. Jang**, Y. Ahn, et al. (2026). Clinically Grounded Retinal Representation Learning from Minimal Supervision. *Scientific Reports*.

[j.6] **B. Jang**, Y. Shin, G. Lee, and Y.-G. Kim (2026). From Static Diagnosis to Dynamic Guidance: Evolution of Artificial Intelligence in Pediatric Neuroimaging. *Journal of Korean Neurosurgical Society*.

- [j.5] J.H. Lee, S.Y. Lee, **B. Jang**, et al. (2026). OCT Biomarkers as Predictors of Treatment Interval in Neovascular Age-Related Macular Degeneration Treated with Intravitreal Aflibercept Using a Treat-and-Extend Regimen. *Scientific Reports*.
- [j.4] **B. Jang**[†], C.H. Lee[†], et al. (2026). Artificial Intelligence-Based Prediction of First Recurrence in Neovascular Age-Related Macular Degeneration with Validation by 19 Experts. *Scientific Reports*.
- [j.3] D.Y. Kim, R. Do, . . . , **B. Jang**, et al. (2025). Automated AI-Based Identification of Autism Spectrum Disorder from Home Videos. *npj Digital Medicine*.
- [j.2] N.Y. Kim[†], **B. Jang**[†], et al. (2024). Differential Diagnosis of Pleural Effusion Using Machine Learning. *Annals of the American Thoracic Society*.
- [j.1] **B. Jang**[†], S.Y. Lee[†], et al. (2023). Preliminary Analysis of Predicting the First Recurrence in Patients with Neovascular Age-Related Macular Degeneration Using Deep Learning. *BMC Ophthalmology*.

PRESENTATIONS

[o] oral presentation; [po] poster presentation

- [o.7] **B. Jang**, G. Lee, J. Choi, and Y.-G. Kim. “Structure-Preserving Test-Time Adaptation for Cross-Modality Retinal Vessel Segmentation.” *2026 Spring Conference of KoSMI*, Cheongju, Korea.
- [po.3] **B. Jang**, K.H. Bae, and Y.-G. Kim. “Assessment of Systemic Health Using Retinal Age Gap: Development of a Predictive Model and Clinical Applicability.” *2025 Annual Conference of the Korea Society of Artificial Intelligence in Medicine (KoSAIM)*, Korea.
- [o.6] **B. Jang** et al. “Assessment of Systemic Health Using Retinal Age Gap: Development of a Predictive Model and Clinical Applicability” *MICCAI 2025 AMAI Workshop*, Daejeon, Korea. **Best Abstract Award Honorable Mention**.
- [o.5] **B. Jang** et al. “Assessment of Systemic Health Using Retinal Age Gap: Development of a Predictive Model and Clinical Applicability.” *2025 Spring Conference of KoSMI*, Busan, Korea. **Best Paper Award**.
- [o.4] **B. Jang** et al. “Self-Supervised Learning for Mesangial Proliferation Prediction in Digital Pathology.” *The 45th Annual Meeting of the Korean Society of Nephrology (KSN 2025)*, Seoul, Korea.
- [o.3] **B. Jang** and Y.-G. Kim. “Foundation Models in Healthcare: Development, Necessity and Implementation Strategies.” *2025 HIMSS Global Health Conference & Exhibition (HIMSS 2025)*, Las Vegas, NV, USA.
- [o.2] **B. Jang** et al. “The Impact of AI-Assistance on Predicting Recurrence in Neovascular Age-Related Macular Degeneration for Ophthalmologists.” *2023 Fall Conference of KoSMI*, Bundang, Korea. **Best Paper Award**.
- [po.2] **B. Jang** et al. “Development and Validation of Pretrained Model for Fundoscopy-Specific using Large-Scale Images.” *MICCAI 2023 OMIA-X Workshop*, Vancouver, Canada.
- [po.1] **B. Jang** et al. “Predicting the First Recurrence in Patients with Neovascular AMD Using Deep Learning.” *2023 American Society of Retina Specialists (ASRS)*, Seattle, USA. **Poster Award Winner**.
- [o.1] **B. Jang** et al. “Development and Validation of Pretrained Model for Fundoscopy-Specific using Large-Scale Images.” *2023 Spring Conference of KoSMI*, Daegu, Korea. **Best Paper Award**.

HONORS & AWARDS

- | | | |
|-------|--|-----------|
| [a.5] | AI Seoul Tech Graduate Scholarship , Seoul Future Talent Foundation. | Nov. 2025 |
| [a.4] | SNUAIMED Excellence Scholarship , Seoul National University. | 2023–2024 |
| [a.3] | Hana Bank President’s Award , 2024 Digital Wellness Competition. | Nov. 2024 |
| [a.2] | 2nd Place , Healthcare AI Learning Platform Challenge 2023, Themes 4 & 6. | Dec. 2023 |
| [a.1] | 3rd Place , Digital Pathology AI Hackathon, CODIPAI Challenge. | Oct. 2023 |

TEACHING & LEADERSHIP

- | | | |
|-------|---|-----------|
| [t.5] | NVIDIA DLI Ambassador and Certified Instructor , NVIDIA Deep Learning Institute. | 2023–2025 |
| [t.4] | Student Ambassador , MICCAI 2025 AMAI Workshop. | Sep. 2025 |
| [t.3] | Teaching Assistant , Clinical and Healthcare AI Training Programs. | 2024–2025 |
| [t.2] | Teaching Assistant , Korea Clinical Datathon (KCD) 2025. | Oct. 2025 |
| [t.1] | Teaching Assistant , NVIDIA MONAI Bootcamp, HCLS Summit Korea 2022. | May 2022 |

PATENTS

- [p.2] Y.-G. Kim, ..., **B. Jang**. *Controllable Normal-to-Disease Fundus Translation via Factorial Guidance*. Korean Patent Application No. 10-2026-0119988, filed June 2026. **Pending**.
- [p.1] H.-S. Kim, ..., Y.-G. Kim, **B. Jang**. *Method and System for Predicting Information Related to Abdominal Surgery from Medical Image Data*. Korean Patent No. 10-2953173, registered April 2026. **Granted**

SKILLS

Programming & Computing: Python, PyTorch, MATLAB; multi-GPU training ($4\times$ A100); Docker, Git, Linux

Methods: Foundation models, representation learning, fine-tuning, multimodal learning, generative modeling, test-time adaptation, domain adaptation, robustness evaluation, representation analysis

Biomedical Data: Fundus, OCT/OCTA, fluorescein angiography, CT, digital pathology, EHR, laboratory data, and health screening data

Tools: Hugging Face, MONAI, PyRadiomics, 3D Slicer

Languages: Korean (native), English (fluent), Chinese (fluent)

REFERENCES

Jinwook Choi, PhD

Professor, Department of Biomedical Engineering, College of Medicine
Seoul National University, Seoul, Republic of Korea
Email: jinchoi@snu.ac.kr

Young-Gon Kim, PhD

Associate Professor, Department of Transdisciplinary Medicine
Seoul National University Hospital, Seoul, Republic of Korea
Email: yg@snuh.org

Kunho Bae, MD, PhD

Professor, Department of Ophthalmology
Seoul National University Hospital, Seoul, Republic of Korea
Mass Eye and Ear, Harvard Medical School, Boston, MA, USA
Email: kunho.bae@snu.ac.kr